SAMUEL SHLYAM

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ACADEMICS

Stevens Institute of Technology – School of Business, Hoboken, NJ

Master of Science in Financial Engineering **September 2026 – May 2027**
GPA: 3.90
Bachelor of Science in Quantitative Finance **September 2023 – May 2026**
GPA: 3.85

Relevant Coursework:

Stochastic Calculus	Statistical Learning	Pricing and Hedging
Financial Statistics	Financial Time Series	Market Microstructure
Optimization for Finance	Financial Risk Management	Algorithmic Trading Strategies
Machine Learning in Finance	Derivatives Pricing	Linear Algebra

Certifications:
Bloomberg Certified, Introduction & Intermediate Python for Finance, Quantitative Risk Management, Introduction to Portfolio Risk Management, Bond Valuation and Analysis in Python

TECHNICAL SKILLS

Programming Languages: Python (NumPy, Pandas, PyTorch), R, C++
Machine Learning: Deep Learning (CNNs, RNNs/LSTMs), Transformers, Reinforcement Learning Fundamentals, Natural Language Processing (LLMs, RAG), Financial Time Series (ARIMA/GARCH)
Quantitative Finance: Monte Carlo Simulation, Stochastic Volatility Modeling (Heston/Bates), Optimization, Mean-Variance Theory

PROJECTS

Sentiment Analysis Augmented Option Pricing **July 2025 – Current**
Awarded "Outstanding Senior Design Project – Research Track"

- Developed a hybrid deep learning framework to estimate parameters for Heston, Bates, and Bergomi models by training a Convolutional Neural Network (CNN) on synthetic volatility surfaces.
- Engineered a secondary architecture that augments the baseline CNN with a synthetic sentiment process to analyze if alternative data improves parameter prediction and pricing accuracy on real-world market surfaces.
- Performed comparative analysis between the baseline model and the sentiment-augmented model to identify economically meaningful improvements in derivative pricing engine performance.

Machine Learning in Finance (FA 690) **January 2026 – Current**

- Developed a modern CNN architecture to classify financial candlestick images and predict subsequent price movements.
- Implemented RNN and LSTM structures to analyze financial news and determine if news sentiment can effectively predict stock price movement.

EXPERIENCE

Icon Luxury Group, New York, New York **December 2023 – September 2025**
Software Development Intern

- Worked alongside the Technical Lead to develop a data mining product for company-wide utility, reducing reliance on data entry and improving data accuracy by more than 40%
- Enhanced data extraction efficiency by 25% through the development of a sophisticated web scraper, utilizing advanced algorithms and multithreading for improved performance
- Transformed raw data into actionable insights, leading the transition from proof-of-concept to production-ready solutions with efficient data processing techniques, achieving a scalable operational framework